Luke Priestley

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Current Role

Postdoctoral Researcher, University of Oxford **Jan 2025–present**

Supervisor: Thomas Akam

Studying the neural mechanisms of planning and decision-making using computational and systems neuroscience approaches.

Education

DPhil, Computational and Cognitive Neuroscience, University of Oxford **2020–24**

MSc, Experimental Psychology, University of Oxford **2018–19**

BA (Hons.) Philosophy and Psychology, University of Sydney **2014–17**

Selected Publications

Priestley, L. and Akam, T. (2026). Dopamine ramps as a normative consequence of dual-process control. *eLife Reviewed Preprint, 111458, version 1.* DOI | Code

Priestley, L., Mahmoodi, A., Reith, W. D., Khalighinejad, N. and Rushworth, M. F. S. (2026). Activity in human dorsal raphe nucleus signals changes in behavioural policy. *Nature Communications, 17, 1665.* DOI | Code

Vogel, T. A., **Priestley, L.**, Cutler, J., Hogg, T., Khalighinejad, N., Garrett, N., Apps, M. A. J., Rushworth, M. F. S. and Lockwood, P. L. (2026). Humans are more prosocial in poor foraging environments. *Nature Communications, 17, 483.* DOI

Priestley, L., Chiew, M., Shahdloo, M., Mahmoodi, A., Cheng, X., Cleveland, R., Rushworth, M. and Khalighinejad, N. (2025). Dorsal raphe nucleus controls motivation-state transitions in monkeys. *Science Advances, 11, eads1236.* DOI

Trier, H. A., Khalighinejad, N., Hamilton, S., Harbison, C., **Priestley, L.**, Laubach, M., Klein-Flügge, M., Scholl, J. and Rushworth, M. F. S. (2025). A distributed subcortical circuit linked to instrumental information-seeking about threat. *Proceedings of the National Academy of Sciences, 122, e2410955121.* DOI

Shahdloo, M., Khalighinejad, N., **Priestley, L.**, Rushworth, M. and Chiew, M. (2024). Dynamic off-resonance correction improves functional image analysis in fMRI of awake behaving non-human primates. *Frontiers in Neuroimaging, 3, 1336887.* DOI

Klein-Flügge, M. C., Jensen, D. E. A., Takagi, Y., **Priestley, L.**, Verhagen, L., Smith, S. M. and Rushworth, M. F. S. (2022). Relationship between nuclei-specific amygdala connectivity and mental health dimensions in humans. *Nature Human Behaviour, 6, 1705–1722.* DOI

Khalighinejad, N.*, Garrett, N.*, **Priestley, L.***, Lockwood, P. and Rushworth, M. F. S. (2021). A habenula-insular circuit encodes the willingness to act. *Nature Communications, 12, 6329.* DOI

Khalighinejad, N., **Priestley, L.**, Jbabdi, S. and Rushworth, M. F. S. (2020). Human decisions about when to act originate within a basal forebrain–nigral circuit. *Proceedings of the National Academy of Sciences, 117, 11799–11810.* DOI

* Joint first authorship.

Scholarships

Clarendon Scholarship, University of Oxford (~£120,000) **2020–24**

Full doctoral funding covering international fees and stipend.

James Fairfax Oxford-Australia Scholarship (~£30,000) **2020–23**

University College Oxford-Australia Scholarship (~£10,000) **2018–19**

J.B. Watt Travelling Scholarship, University of Sydney (~£10,000) **2018–19**

Other Awards

Senior Hulme Scholar, Brasenose College, University of Oxford **2022–24**

Awarded for outstanding academic distinction among doctoral candidates.

Oldfield Prize, Department of Experimental Psychology, University of Oxford **2019**

Awarded for best academic performance in the MSc programme.

University Medal, University of Sydney **2017**

Awarded for outstanding academic achievement across undergraduate and honours study.

O'Neil Prize, University of Sydney **2017**

Awarded for best honours research thesis.

Academic Merit Prize, University of Sydney **2014–17**

Awarded for exceptional undergraduate academic performance.

Supervision

Annika Zhang, University of Oxford **2023–24**

MSc in Experimental Psychology. Now PhD candidate at King's College London.

William Reith, University of Oxford **2022–23**

MSc in Neuroscience, University of Oxford. Now PhD candidate at

Sainsbury Wellcome Centre for Neural Circuits and Behaviour, University College London.

Zoe Hilker, University of Oxford **2021–22**

MSc in Experimental Psychology, University of Oxford. Now PhD candidate at Max Planck Centre for Cognition, Leipzig.

Teaching

2nd Year Behavioural Neuroscience, University of Oxford **2021–23**

Co-organiser and demonstrator.

3rd Year Neuroanatomy, University of Oxford **2021–23**

Course demonstrator.

Invited Talks

Oxford Neuroscience Symposium **2024**
Dorsal Raphe Nucleus Controls Motivation-State Transitions in Monkeys.

BrainBox **2024**
Dorsal Raphe Nucleus Controls Motivation-State Transitions in Monkeys.

University of Sydney (Shine & O'Callaghan groups) **2024**
A Role for Dorsal Raphe Nucleus in Human and Non-human Primate Behavioural Control.

Donders Institute (den Ouden group) **2023**
Dorsal Raphe Nucleus Controls Motivation-State Transitions in Monkeys.

Conference Presentations

COSYNE (poster) **2026**
Dopamine Ramps as a Normative Consequence of Dual-Process Control.

Conference on Computational Cognitive Neuroscience (poster) **2023**
Dorsal Raphe Nucleus Controls Motivation-State Transitions in Monkeys.

Symposium on the Biology of Decision Making (poster) **2023**
Dorsal Raphe Nucleus Controls Motivation-State Transitions in Monkeys.